Solution Key: Probability Readiness / Waiver Test

for a Berkeley-style Stat 134 course

1.
$$\mathbb{P}(H = 6 \mid H \geq 5) = \frac{\binom{10}{6}}{\binom{10}{5} + \binom{10}{6} + \binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10}} = \frac{210}{638} = \frac{105}{319}.$$

2. Let D be disease and $+$ positive test. Then
$$\mathbb{P}(D \mid +) = \frac{0.97 \cdot 0.01}{0.97 \cdot 0.01 + 0.04 \cdot 0.99} = \frac{0.0097}{0.0493} \approx 0.1968.$$

3. Example: take sample space $\Omega = \{00, 01, 10, 11\}$ with uniform probability, and define
$$A = \{00, 01\}, \quad B = \{00, 10\}, \quad C = \{00, 11\}.$$
Then any two are independent, but
$$\mathbb{P}(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C).$$

4. Let I_j be the indicator that the jth draw is red. Then $X = I_1 + I_2 + I_3$ and
$$\mathbb{E}[X] = \sum_{j=1}^{3} \mathbb{E}[I_j] = 3 \cdot \frac{7}{12} = \frac{7}{4}.$$
By symmetry, each draw has marginal probability 7/12 of being red.

5. Since $\mathbb{P}(X > k) = (1-p)^k$,
$$\mathbb{P}(X > m + n \mid X > m) = \frac{\mathbb{P}(X > m + n)}{\mathbb{P}(X > m)} = \frac{(1-p)^{m+n}}{(1-p)^m} = (1-p)^n = \mathbb{P}(X > n).$$

6. Since $Y = X^2$ and $X \in (0, 1)$, we have $X = \sqrt{y}$ and
$$f_Y(y) = f_X(\sqrt{y}) \left| \frac{d}{dy} \sqrt{y} \right| = 2\sqrt{y} \cdot \frac{1}{2\sqrt{y}} = 1, \qquad 0 < y < 1.$$
So $Y \sim \text{Unif}(0, 1)$.

7. For $0 < x < 1$,
$$f_X(x) = \int_0^x 2\, dy = 2x.$$
Also
$$f_{Y|X}(y \mid x) = \frac{f(x, y)}{f_X(x)} = \frac{2}{2x} = \frac{1}{x}, \qquad 0 < y < x.$$
Thus $Y \mid X = x \sim \text{Unif}(0, x)$ and
$$\mathbb{E}[Y \mid X = x] = \frac{x}{2}.$$

8. By the law of total expectation,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid B]] = \frac{2}{3} \cdot 0 + \frac{1}{3} \cdot 2 = \frac{2}{3}.$$

For the variance,

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid B)] + \mathrm{Var}(\mathbb{E}[X \mid B]).$$

The first term is 1. The second is the variance of a variable taking values 0 and 2 with probabilities 2/3 and 1/3:

$$\mathrm{Var}(\mathbb{E}[X \mid B]) = \frac{2}{3}(0 - 2/3)^2 + \frac{1}{3}(2 - 2/3)^2 = \frac{8}{9}.$$

Hence

$$\mathrm{Var}(X) = 1 + \frac{8}{9} = \frac{17}{9}.$$

9. Use the identity

$$X = \sum_{k=1}^{\infty} \mathbf{1}_{\{X \geq k\}}.$$

Taking expectations and interchanging sum and expectation gives

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k).$$

10.

$$\mathrm{Var}(\overline{X}_n) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{\sigma^2}{n},$$

using independence.

11. If $X_1, X_2, \ldots$ are i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$, then

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} Z, \qquad Z \sim \mathcal{N}(0,1).$$

Equivalently,

$$\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \xrightarrow{d} Z.$$

12. Here $\mu = np = 80$ and $\sigma^2 = np(1-p) = 48$, so $\sigma = \sqrt{48}$. With continuity correction,

$$\mathbb{P}(S_{200} \leq 70) \approx \Phi\left(\frac{70.5 - 80}{\sqrt{48}}\right).$$

13. Here $\lambda = np = 4000(0.0005) = 2$. Hence

$$\mathbb{P}(X = 1) \approx e^{-2}\frac{2^1}{1!} = 2e^{-2}.$$

14. The waiting time to the third arrival is the sum of three independent $\mathrm{Exp}(\lambda)$ variables, hence has Gamma distribution with shape 3 and rate λ. Its density is

$$f(t) = \frac{\lambda^3 t^2 e^{-\lambda t}}{2}, \qquad t > 0.$$

15. Conditional on $N(5) = 3$, the three arrival times are distributed as the order statistics of three independent $\mathrm{Unif}(0, 5)$ random variables. Equivalently, the unordered arrival times are i.i.d. $\mathrm{Unif}(0, 5)$.

16. Let $\pi = (\pi_0, \pi_1)$. Solve
$$\pi_0 = 0.8\pi_0 + 0.3\pi_1, \qquad \pi_0 + \pi_1 = 1.$$
Thus $0.2\pi_0 = 0.3\pi_1$, so $2\pi_0 = 3\pi_1$. Hence
$$\pi_0 = \frac{3}{5}, \qquad \pi_1 = \frac{2}{5}.$$

17. For fair gambler's ruin on $\{0, 1, 2, 3, 4, 5\}$,
$$\mathbb{P}_i(\tau_5 < \tau_0) = \frac{i}{5}.$$
So from $i = 2$ the answer is
$$\frac{2}{5}.$$

18. For the fair absorbed walk on $\{0, 1, \ldots, N\}$,
$$\mathbb{E}_i[T] = i(N - i).$$
Here $N = 5$ and $i = 2$, so
$$\mathbb{E}_2[T] = 2(5 - 2) = 6.$$

19. The offspring pgf is
$$f(s) = 0.4 + 0.6s^2.$$
The extinction probability q satisfies
$$q = 0.4 + 0.6q^2.$$
This gives
$$0.6q^2 - q + 0.4 = 0.$$
The roots are $q = 1$ and $q = 2/3$. The extinction probability is the smaller root in $[0, 1]$, hence
$$q = \frac{2}{3}.$$

20. Let $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$. Since
$$S_{n+1} = S_n + X_{n+1},$$
we have
$$S_{n+1}^2 = S_n^2 + 2S_nX_{n+1} + X_{n+1}^2.$$
Taking conditional expectation given $\mathcal{F}_n$,
$$\mathbb{E}[S_{n+1}^2 - (n+1) \mid \mathcal{F}_n] = S_n^2 + 2S_n\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] + \mathbb{E}[X_{n+1}^2 \mid \mathcal{F}_n] - (n+1).$$
Now $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = 0$ and $X_{n+1}^2 = 1$, so this becomes
$$S_n^2 + 0 + 1 - (n+1) = S_n^2 - n = M_n.$$
Hence M_n is a martingale.